

Quick Guide for GIWAXS_sim

The first cell is to calculate the structure factors by using MDAnalysis package.

```
# Load pdb structure file
pdb_path = r"C:\Users\wax20\Downloads\giwaxs_sim\pdb\Figure_S5b.pdb"
univ = mda.Universe(pdb_path)
transform = univ.trajectory.add_transformations(wrap(univ.atoms))
atoms = univ.select_atoms("not name V")
pos = atoms.positions.copy() # (N atoms, 3)
```

In pdb_path to load the file path of the pdb file.

```
# Extract top sturcture factors, use # around 15 +~2 for top15, ~100 for top100
top100 = res_filtered[:15]

print("Top 15 Structure Factor (h,k,l): Gr, Gz, S")
for entry in top100:
    print(entry)

Computing S(G): 100% | ████████████████████████████████████████ | 285483/285483 [05:16<00:00, 902.88it/s]
```

Top 15 Structure Factor (h,k,l): Gr, Gz, S

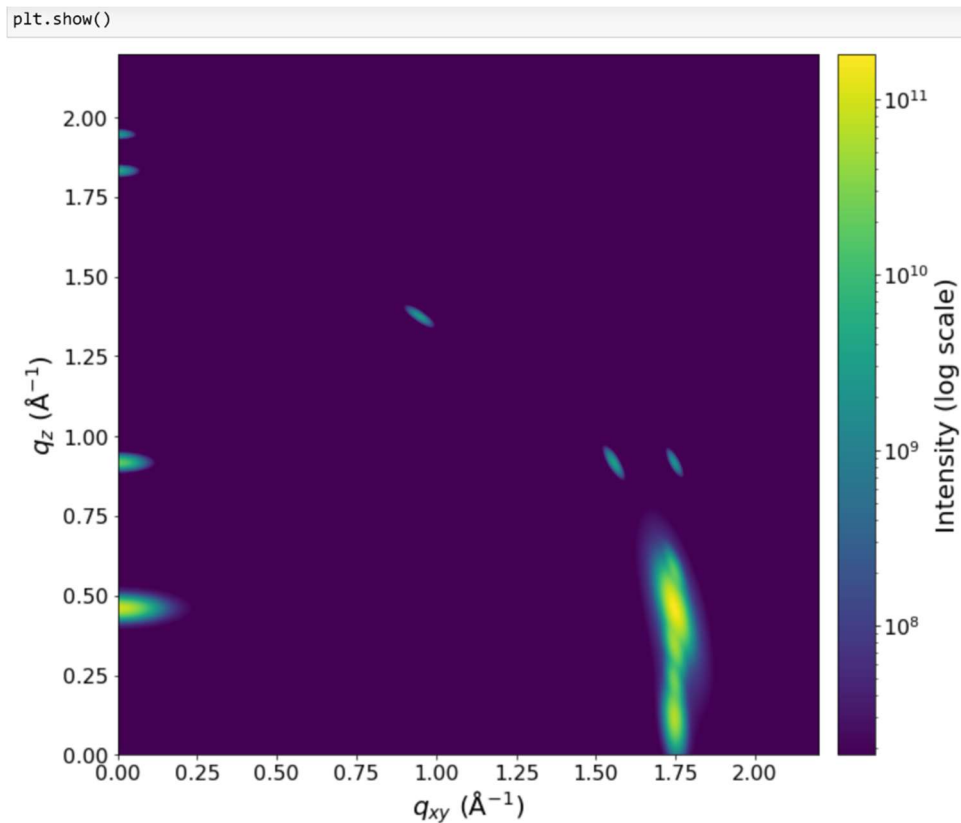
- (0, -12, 4, 1.80501139, 0.45785801, 1.72558507e+11)
- (0, 0, 4, 0.45785801, 0.45785801, 9.43521337e+10)
- (0, -12, 1, 1.74972405, 0.1144645, 6.62936221e+10)
- (0, -12, 3, 1.77942441, 0.34339351, 5.05354457e+10)
- (0, -12, 2, 1.76092039, 0.22892901, 2.77144764e+10)
- (0, 0, 8, 0.91571602, 0.91571602, 1.99880731e+10)
- (0, -12, 5, 1.83738543, 0.57232252, 1.84387066e+10)
- (-1, -12, 4, 1.80513334, 0.45785801, 6.79084245e+09)
- (0, 0, 16, 1.83143205, 1.83143205, 4.64822722e+09)
- (-40, -9, 8, 1.80491591, 0.91571602, 3.92094712e+09)
- (-2, -12, 4, 1.80549916, 0.45785801, 3.86060881e+09)
- (-40, -3, 12, 1.66783547, 1.37357404, 3.11114653e+09)
- (-1, -12, 3, 1.77954812, 0.34339351, 2.91980746e+09)
- (0, 0, 17, 1.94589655, 1.94589655, 2.6824797e+09)
- (0, -12, 8, 1.9715395, 0.91571602, 2.38232574e+09)

Change the number in `res_filtered[:]` to determine the structure factors of top intensities in the structure and run the first cell to compute $S(G)$ (take a couple mins).

In this work, we calculated around top 15 $S(G)$ for the crystalline structures and top 100 $S(G)$ for the amorphous structures.

Note that if you changed the code “top100” to “top15”, then you need to change all the following “top100” in the rest of the code including in the second cell. Just change the # in the res_filtered[:] is fine.

After the computation is finished, just run the second cell will display the simulated GIWAXS pattern for our coarse grain model structures.



```
# Function to draw the (h,k,L) labels
...
def adjust_labels(ax, xs, ys, labels):
    for x, y, label in zip(xs, ys, labels):
        ax.annotate(
            f"{label}",
            xy=(x, y),
            xytext=(10, 10),          # offset in points
            textcoords='offset points',
            fontsize=8,
            color='white',           # white text for contrast
            ha='left',
            va='bottom'
        )
    adjust_labels(ax, qr, Gz_vals, hkl_vals)
...
```

Activate this part of the second cell will show the plane label on the pattern.